



Programme syllabus

[An accessible version of the syllabus can be found in the Course and programme directory.](#)

Master's Programme, Applied and Computational Mathematics 120.0 credits

Masterprogram, tillämpad matematik och beräkningsmatematik

Valid for students admitted to the education from autumn 26 (HT - Autumn term; VT - Spring term).

This is a translation of the Swedish, legally binding, programme syllabus.

Programme objectives

The main objective of this programme is to train skilled applied mathematicians, well prepared for advanced industrial positions or continued graduate studies.

Knowledge and understanding

A Master of Science in Applied and Computational Mathematics shall:

- demonstrate knowledge and understanding in Applied Mathematics, including both broad knowledge of the field and a considerable degree of specialised knowledge within the chosen area of specialisation as well as insight into current research and development work,
- demonstrate specialised methodological knowledge in Applied Mathematics, including techniques of mathematical modelling, analysis of mathematical models, and simulation.
- be able to formulate and approach new problem settings in a scientific manner, by having a creative, critical and systematic attitude towards applied mathematics.

Skills and abilities

A Master of Science in Applied and Computational Mathematics shall be able to:

- demonstrate the ability to critically and systematically integrate knowledge and analyse, assess and deal with complex phenomena, issues and situations even with limited information,
- demonstrate the ability to identify and formulate issues critically, autonomously and creatively as well as to plan and, using appropriate methods, undertake advanced tasks within predetermined time frames and so contribute to the formation of knowledge as well as the ability to evaluate this work,
- demonstrate the ability in speech and writing both nationally and internationally to clearly report and discuss his or her conclusions and the knowledge and arguments on which they are based in dialogue with different audiences,
- demonstrate the skills required for participation in research and development work or autonomous employment in some other qualified capacity,
- independently apply mathematical theories, methods, and models,
- formulate mathematical models, select suitable methods to investigate those models including the efficient use of computer tools.

Ability to make judgments and adopt a standpoint

A Master of Science in Applied and Computational Mathematics shall be able to:

- demonstrate the ability to make assessments in Applied Mathematics informed by relevant disciplinary, social and ethical issues and also to demonstrate awareness of ethical aspects of research and development work,
- demonstrate insight into the possibilities and limitations of research, its role in society and the responsibility of the individual for how it is used,
- critically judge the validity and limitations of results obtained from different types of mathematical models,

- demonstrate the ability to identify the personal need for further knowledge and take responsibility for his or her ongoing learning.

Extent and content of the programme

Applied and Computational Mathematics is a two-year (120 university credits) Master's programme at the advanced level (second cycle). The instruction language is entirely English. The programme consists of a basic curriculum followed by four tracks: (i) computational mathematics, (ii) optimization and systems theory, (iii) mathematics of data science, and (iv) financial mathematics. The courses in the basic curriculum are compulsory and worth 30 university credits. To obtain sufficient depth in a track, a student is required to complete courses worth approximately 30 university credits among the profile courses for the track in question. In addition to these four tracks there is a fifth track (v) computer simulations for science and engineering. This track is given according to what is stated further in this syllabus. The language of instruction is English. Language courses may be given in Swedish, German or Dutch.

The track CSSE includes compulsory mobility for the students. The track includes three semesters of courses followed by the fourth research semester spent on the Master's Thesis (30 ECTS) under the supervision of both degree-awarding universities. The joint supervision of the thesis work includes the possibility that the topic is selected according to specializations offered by the degree-awarding universities individually. As a general rule, students will be admitted to TU Berlin for the 1st year, for course work amounting to at least 60 ECTS as outlined in Annex 1. Upon request, students may enter KTH or TU Delft for their 1st year of studies (course work amounting to at least 60 ECTS as outlined in Annex 1), if approved by the Programme Advisory Group (PAG) and endorsed by the said university. Admission for the 2nd year can in this case be either TU Berlin or KTH/TU Delft as approved by PAG and endorsed by the university being responsible for admitting the student to the 2nd year.

The second year university offers course work (at least 30 ECTS) and supervision of the Master's thesis (30 ECTS) as outlined in Annex 1. This includes a choice of a specialization at the chosen university:

KTH: Biocomputing, bioinformatics, machine learning, computational fluid dynamics, numerical linear algebra, computational physics, numerical analysis;

TU Delft: computational fluid dynamics, numerical linear algebra, high-performance computing, data assimilation

TU Berlin: Optimal Control, Control Theory, Numerical Linear Algebra, Numerical Analysis

A student may select a specialisation different from the proposals above after a separate application and subsequent approval by the programme director.

Eligibility and selection

General eligibility requirements

- A completed Bachelor's degree, corresponding to a Swedish Bachelor's degree (180 ECTS), or equivalent academic qualifications from an internationally recognised university. Students in their final year of undergraduate education may also apply to KTH and if qualified, receive a conditional acceptance.
- English language proficiency equivalent to (the Swedish upper secondary school) English course B/6. There are different ways to fulfill the English language requirements, see: www.kth.se

Specific eligibility requirements

A Bachelor's degree, or equivalent, corresponding to 180 ECTS credits, with courses and/or course modules in:

Mathematics and Programming: equivalent to approximately 45 ECTS credits in total which must include courses or course modules on

- partial differential equations
- numerical methods
- mathematical statistics
- programming

Selection process

The selection process is based on the following criteria:

Study results (grades, relevant subjects and English), motivation for studies (motivation letter) and prior education. Prior education is evaluated based on the quality and suitability of the subjects that are relevant for the applied program. The merit rating is carried out on a scale of 1-75.

Implementation of the education

Structure of the education

The duration of the academic year at KTH is 40 weeks. The academic year is divided into four periods. Each period lasts approximately seven weeks with at least 33 days of study. Each period is followed by an examination period. In addition to the four regular examination periods, there are four re-examination periods: at Christmas, in March, June and August, immediately before the start of the academic year. Teaching activities may, if necessary, be scheduled outside the academic year. Details about the structure follow the general rules stated by KTH.

The first year of the track CSSE is carried out at one university and is devoted to the compulsory core courses and the preparation courses for the specialization that the student chooses. The choice of specialization is done at latest after the first year when moving to the second university. The second year is carried out at a different university than the first. The studies comprise specialization courses, elective courses and a degree project.

Courses

The programme is course-based. Lists of courses are included in appendix 1.

At least 90 university credits worth of courses must be taken. Of these, courses worth 30 credits are mandatory for all students of the programme. For each track there is an additional ca 30 credits worth of courses that are mandatory for that track. The remaining ca 30 credits are obtained by taking optional courses. The optional courses can be chosen freely if they are at the advanced level given by the department of mathematics, or if they have a strong connection to applied mathematics. A list of such optional courses is included in appendix 1. To some extent, other choices of courses are also possible. In particular, upto 15 credits can typically be non-technical ("teknikkomplementära") courses.

Note that the choice of optional courses must always be approved by the programme director.

Grading system

Courses in the first and the second cycle are graded on a scale from A to F. A-E are passing grades, A is the highest grade. The grades pass (P) and fail (F) are used for courses under certain circumstances.

Courses in the first and the second cycle are graded on a scale from A to F. A-E are passing grades, A is the highest grade. The grades pass (P) and fail (F) are used for courses under certain circumstances.

The grades pass (P) and fail (F) are used for thesis works.

Conditions for participation in the programme

Before the second semester each student must select one of the four tracks: computational mathematics, optimization and systems theory, statistical learning and data analytics or financial mathematics. At least 45 credits worth of course work must have been finished during the first year (including the re-examination period in August) to be allowed to move on to the second year of the programme.

For the track CSSE the student is required to make a course selection for the coming semester. This must be done according to the rules of the university where the studies will be carried out. The student is responsible for having the recommended prerequisites. In a few courses, the number of places is limited and selection, for the students who applied in time, is done according to merits such as grades and credits. The selection is done at the course-giving university. For each course, the student shall, at the start of course, register for the course according to the rules of the university giving the course. For students starting their education from the autumn semester 2018, previous promotion requirements have been replaced with special admission requirements to each course. Admission requirements are specified in the course syllabus.

For students starting their education from the autumn semester 2018, previous promotion requirements have been replaced with special admission requirements to each course. Admission requirements are specified in the course syllabus.

Course application

As a student at KTH programmes you have to apply for semester courses. The application is done via www.universityadmissions.se

Course registration

Students admitted to an educational programme at KTH must register for the courses they intend to study. Course registration is required for the examination and means that the student is active.

Degree project

A 30-credit Master's degree project is carried out at the end of the educational programme (usually the fourth semester). The purpose of the degree project is to develop the student's ability to perform independent project work, as well as to apply and deepen the skills obtained from the courses of the programme. The project may be carried out in an academic or industrial environment in Sweden or abroad.

The student must actively search for a suitable degree project; however KTH can provide some assistance with information on suitable points of contact. The choice of project must be approved by the master thesis responsible for the selected track.

To be allowed to start a degree project, a student must have accumulated at least 60 credits of the total course work, and enough depth within the selected track.

The degree project is graded with P/F.

Students within the track CSSE must complete an individual degree project which corresponds to 30 ECTS credits. In order to start the degree project, at least 60 ECTS credits of the total number of course credits must be completed, where 10 ECTS credits are from the student's specialization. The goal with the degree project is that the student shall show the ability to independently carry out a degree project by using the knowledge and abilities acquired throughout the programme. The degree project is carried out at the university responsible for the student's second year and is supervised by teachers from both universities.

Degree

A Degree of Master of Science is obtained after completing the degree programme.

Passing grades in all courses which are included in the student's study plan are required. The study plan must comprise 120 higher education credits which include a degree project consisting of 30 higher education credits.

KTH's local degree ordinance can be found at KTH's website, www.kth.se.

In order for a student at track CSSE to receive double degrees from both universities, all courses in the individual study plan must be passed and the degree project must be deemed as passed by both the home and host university. For CSSE students who have studied one of the two years at KTH, it is required that the student has fulfilled the degree ordinance and has completed courses comprising 60 ECTS credits at KTH. The degree project, carried out at the host university, must correspond to 30 ECTS credits. Students who fulfil all requirements are awarded the Degree of Master of Science.

Application for degree certificate

When the studies at KTH are completed the student can apply for a degree certificate. Application is done through the "personal menu" at www.kth.se

Students at the track CSSE can apply for a degree certificate from both the universities where they have studied.

Appendix 1: Course list

Appendix 2: Specialisations



Appendix 1: Course list

Master's Programme, Applied and Computational Mathematics (TTMAM)

General Courses

Study year 1

Mandatory courses (15.0 credits)

Course code	Course name	Credits	Edu. level
AK2030	Theory and Methodology of Science (Natural and Technological Science)	4.5 credits	Second cycle
SA2001	Sustainable development and research methodology in mathematics	3.0 credits	Second cycle
SF2940	Probability Theory	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2812	Applied Linear Optimization	7.5 credits	Second cycle
SF2524	Matrix Computations for Large-scale Systems	7.5 credits	Second cycle
SF2527	Numerical Methods for Differential Equations I	7.5 credits	Second cycle
SF2832	Mathematical Systems Theory	7.5 credits	Second cycle
SF2863	Systems Engineering	7.5 credits	Second cycle

Optional courses

Course code	Course name	Credits	Edu. level
DD2356	Methods in High Performance Computing	7.5 credits	Second cycle
DD2365	Advanced Computation in Fluid Mechanics	7.5 credits	Second cycle
DD2421	Machine Learning	7.5 credits	Second cycle
SF1677	Foundations of Analysis	7.5 credits	First cycle
SF1678	Groups and Rings	7.5 credits	First cycle
SF1691	Complex Analysis	7.5 credits	First cycle
SF2526	Numerical algorithms for data-intensive science	7.5 credits	Second cycle
SF2701	Financial Mathematics, Basic Course	7.5 credits	Second cycle
SF2822	Applied Nonlinear Optimization	7.5 credits	Second cycle
SF2842	Geometric Control Theory	7.5 credits	Second cycle
SF2930	Regression Analysis	7.5 credits	Second cycle
SF2943	Time Series Analysis	7.5 credits	Second cycle
SF2955	Computer Intensive Methods in Mathematical Statistics	7.5 credits	Second cycle
SF2971	Martingales and Stochastic Integrals	7.5 credits	Second cycle
SG2212	Computational Fluid Dynamics	7.5 credits	Second cycle
SG2224	Applied Computational Fluid Dynamics	5.0 credits	Second cycle
DD2601	Deep Generative Models and Synthesis	7.5 credits	Second cycle
DD2257	Visualization	7.5 credits	Second cycle
DD2434	Machine Learning, Advanced Course	7.5 credits	Second cycle
DD2435	Mathematical Modelling of Biological Systems	9.0 credits	Second cycle
DD2440	Advanced Algorithms	6.0 credits	Second cycle
SF1811	Optimization	6.0 credits	First cycle
SF2852	Optimal Control Theory	7.5 credits	Second cycle
SF2866	Applied Systems Engineering	7.5 credits	Second cycle
SF2935	Modern Methods of Statistical Learning	7.5 credits	Second cycle
SF2942	Portfolio Theory and Risk Management	7.5 credits	Second cycle
SF2956	Topological Data Analysis	7.5 credits	Second cycle
SF2975	Financial Derivatives	7.5 credits	Second cycle
SF2980	Risk Management	7.5 credits	Second cycle
DD2352	Algorithms and Complexity	7.5 credits	Second cycle

Supplementary information

At least one of the conditionally elective courses SF2832, SF2863 and SF2812 among the general courses has to be studied, and also minimum one of the courses SF2527 and SF2524.

The conditionally elective courses can be studied during the first or second year.

Students from CTMAT who has taken SF1693 cannot take SF2527, and can choose to take SF2524 or not.

Mandatory SF2940 can be replaced by SF2944 (note: only one can be taken within the TTMAM programme).

Study year 2

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2524	Matrix Computations for Large-scale Systems	7.5 credits	Second cycle
SF2527	Numerical Methods for Differential Equations I	7.5 credits	Second cycle
SF2812	Applied Linear Optimization	7.5 credits	Second cycle
SF2832	Mathematical Systems Theory	7.5 credits	Second cycle
SF2863	Systems Engineering	7.5 credits	Second cycle

Optional courses

Course code	Course name	Credits	Edu. level
DD2257	Visualization	7.5 credits	Second cycle
DD2421	Machine Learning	7.5 credits	Second cycle
DD2434	Machine Learning, Advanced Course	7.5 credits	Second cycle
DD2435	Mathematical Modelling of Biological Systems	9.0 credits	Second cycle
DD2440	Advanced Algorithms	6.0 credits	Second cycle
DD2445	Complexity Theory	7.5 credits	Second cycle
SF1811	Optimization	6.0 credits	First cycle
SF2565	Program Construction in C++ for Scientific Computing	7.5 credits	Second cycle
SF2852	Optimal Control Theory	7.5 credits	Second cycle
SF2866	Applied Systems Engineering	7.5 credits	Second cycle
SF2935	Modern Methods of Statistical Learning	7.5 credits	Second cycle
SF2942	Portfolio Theory and Risk Management	7.5 credits	Second cycle
SF2956	Topological Data Analysis	7.5 credits	Second cycle
SF2957	Statistical Machine Learning	7.5 credits	Second cycle
SF2975	Financial Derivatives	7.5 credits	Second cycle
SF2980	Risk Management	7.5 credits	Second cycle
DD2601	Deep Generative Models and Synthesis	7.5 credits	Second cycle

Supplementary information

At least one of the conditionally elective courses SF2832, SF2863 and SF2812 among the general courses has to be studied, and also minimum one of the courses SF2527 and SF2524.

The conditionally elective courses can be studied during the first or second year.

Students from CTMAT who has taken SF1693 cannot take SF2527, and can choose to take SF2524 or not.

Mandatory SF2940 can be replaced by SF2944 (note: only one can be taken within the TTMAM programme).

Track, Computational Mathematics (COMA)

Study year 1

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2526	Numerical algorithms for data-intensive science	7.5 credits	Second cycle
SF2525	Computational Methods for Stochastic Differential Equations and Machine Learning	7.5 credits	Second cycle
SF2528	Numerical Methods for Differential Equations II	7.5 credits	Second cycle
DD2365	Advanced Computation in Fluid Mechanics	7.5 credits	Second cycle

Information regarding conditionally elective courses

Minimum 4 conditionally elective trackcourses must be studied during year 1 and 2.

Study year 2

Mandatory courses (30.0 credits)

Course code	Course name	Credits	Edu. level
SF250X	Degree Project in Scientific Computing, Second Cycle	30.0 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2524	Matrix Computations for Large-scale Systems	7.5 credits	Second cycle
SF2527	Numerical Methods for Differential Equations I	7.5 credits	Second cycle
SF2529	Inverse Problems	7.5 credits	Second cycle
SF2565	Program Construction in C++ for Scientific Computing	7.5 credits	Second cycle

Information regarding conditionally elective courses

Minimum 4 conditionally elective trackcourses must be studied during year 1 and 2.

Track, Financial Mathematics (FMIA)

Study year 1

Mandatory courses (7.5 credits)

Course code	Course name	Credits	Edu. level
SF2701	Financial Mathematics, Basic Course	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2930	Regression Analysis	7.5 credits	Second cycle
SF2971	Martingales and Stochastic Integrals	7.5 credits	Second cycle
SF2943	Time Series Analysis	7.5 credits	Second cycle

Information regarding conditionally elective courses

At least two conditionally elective courses has to be studied on the track. *One of the conditionally elective courses on the track has to be SF2930 or SF2943.

Study year 2

Mandatory courses (37.5 credits)

Course code	Course name	Credits	Edu. level
SF291X	Degree Project in Financial Mathematics, Second Cycle	30.0 credits	Second cycle
SF2942	Portfolio Theory and Risk Management	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2975	Financial Derivatives	7.5 credits	Second cycle
SF2980	Risk Management	7.5 credits	Second cycle

Information regarding conditionally elective courses

At least two conditionally elective courses has to be studied on the track. *One of the conditionally elective courses on the track has to be SF2930 or SF2943.

Track, Mathematics of Data Science (DAVE)

Study year 1

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2526	Numerical algorithms for data-intensive science	7.5 credits	Second cycle
SF2930	Regression Analysis	7.5 credits	Second cycle
SF2955	Computer Intensive Methods in Mathematical Statistics	7.5 credits	Second cycle
SF2525	Computational Methods for Stochastic Differential Equations and Machine Learning	7.5 credits	Second cycle
SF2943	Time Series Analysis	7.5 credits	Second cycle

Information regarding conditionally elective courses

One mandatory course within the track (7,5cr) and at least three conditionally elective courses within the track (22,5cr) = 30cr.

Study year 2

Mandatory courses (37.5 credits)

Course code	Course name	Credits	Edu. level
SF290X	Degree Project in Mathematical Statistics, Second Cycle	30.0 credits	Second cycle
SF2935	Modern Methods of Statistical Learning	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2529	Inverse Problems	7.5 credits	Second cycle
SF2956	Topological Data Analysis	7.5 credits	Second cycle
SF2957	Statistical Machine Learning	7.5 credits	Second cycle

Information regarding conditionally elective courses

One mandatory course within the track (7,5cr) and at least three conditionally elective courses within the track (22,5cr) = 30cr.

Track, Optimization and Systems Theory (OPST)

Study year 1

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2822	Applied Nonlinear Optimization	7.5 credits	Second cycle
SF2866	Applied Systems Engineering	7.5 credits	Second cycle
SF2832	Mathematical Systems Theory	7.5 credits	Second cycle
SF2863	Systems Engineering	7.5 credits	Second cycle
SF2812	Applied Linear Optimization	7.5 credits	Second cycle
SF2842	Geometric Control Theory	7.5 credits	Second cycle

Study year 2

Mandatory courses (30.0 credits)

Course code	Course name	Credits	Edu. level
SF280X	Degree Project in Optimization and Systems Theory, Second Cycle	30.0 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
SF2832	Mathematical Systems Theory	7.5 credits	Second cycle
SF2852	Optimal Control Theory	7.5 credits	Second cycle
SF2863	Systems Engineering	7.5 credits	Second cycle
SF2866	Applied Systems Engineering	7.5 credits	Second cycle

Information regarding conditionally elective courses

At least 30 cr of the conditionally elective courses need to be studied in the field optimization and systems theory.

Track, Computer Simulations for Science and Engineering (CSSE)

Study year 1

Mandatory courses (22.5 credits)

Course code	Course name	Credits	Edu. level
AK2030	Theory and Methodology of Science (Natural and Technological Science)	4.5 credits	Second cycle
SA2001	Sustainable development and research methodology in mathematics	3.0 credits	Second cycle
SF2524	Matrix Computations for Large-scale Systems	7.5 credits	Second cycle
SF2940	Probability Theory	7.5 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
DD2257	Visualization	7.5 credits	Second cycle
SF2526	Numerical algorithms for data-intensive science	7.5 credits	Second cycle
SF2822	Applied Nonlinear Optimization	7.5 credits	Second cycle
SF2525	Computational Methods for Stochastic Differential Equations and Machine Learning	7.5 credits	Second cycle

Information regarding conditionally elective courses

Conditionally elective course shall be chosen after agreement with the director of the programme.

In the first study year, out of the conditionally elective courses, 3 courses must be selected. In the second study year, at least 30 credits out of the conditionally elective courses must be selected.

Study year 2

Mandatory courses (30.0 credits)

Course code	Course name	Credits	Edu. level
SF259X	Degree Project in Scientific Computing, Second Cycle	30.0 credits	Second cycle

Conditionally elective courses

Course code	Course name	Credits	Edu. level
AK2030	Theory and Methodology of Science (Natural and Technological Science)	4.5 credits	Second cycle
BB2280	Molecular Modeling	7.5 credits	Second cycle
CB2442	Bioinformatics	7.5 credits	Second cycle
DD2421	Machine Learning	7.5 credits	Second cycle
DD2434	Machine Learning, Advanced Course	7.5 credits	Second cycle
DD2435	Mathematical Modelling of Biological Systems	9.0 credits	Second cycle
SA2001	Sustainable development and research methodology in mathematics	3.0 credits	Second cycle
SF2524	Matrix Computations for Large-scale Systems	7.5 credits	Second cycle
SF2565	Program Construction in C++ for Scientific Computing	7.5 credits	Second cycle
SF2832	Mathematical Systems Theory	7.5 credits	Second cycle
SF2863	Systems Engineering	7.5 credits	Second cycle
SF2935	Modern Methods of Statistical Learning	7.5 credits	Second cycle
SF2942	Portfolio Theory and Risk Management	7.5 credits	Second cycle
SF2956	Topological Data Analysis	7.5 credits	Second cycle
SF2980	Risk Management	7.5 credits	Second cycle
SK2532	Biomedicine for Engineers	7.5 credits	Second cycle



Appendix 2: Specialisations

Master's Programme, Applied and Computational Mathematics (TTMAM)

Track, Computational Mathematics (COMA)

The field of computer simulations is of great importance for high-tech industry and scientific/engineering research, e.g. virtual processing, climate studies, fluid dynamics, advanced materials, etc. Thus, **Computational Science and Engineering** (CSE) is an enabling technology for scientific discovery and engineering design. CSE involves mathematical modeling, numerical analysis, computer science, high-performance computing and visualization. The remarkable development of large scale computing in the last decades has turned CSE into the "third pillar" of science, complementing theory and experiment.

The track **Computational Mathematics (COMA)** is mainly concerned with the mathematical foundations of CSE. However, in this track we will also discuss issues of high-performance computing. Given the interdisciplinarity, your final curriculum may vary greatly depending on your interests.

Track, Financial Mathematics (FMIA)

Financial mathematics is applied mathematics used to analyze and solve problems related to financial markets. Any informed market participant would exploit an opportunity to make a profit without any risk of loss. This fact is the basis of the theory of arbitrage-free pricing of derivative instruments. Arbitrage opportunities exist but are rare. Typically both potential losses and gains need to be considered. Hedging and diversification aim at reducing risk. Speculative actions on financial markets aim at making profits. Market participants have different views of the future market prices and combine their views with current market prices to take actions that aim at managing risk while creating opportunities for profits. Portfolio theory and quantitative risk management present theory and methods that form the theoretical basis of market participants' decision making.

Financial mathematics has received lots of attention from academics and practitioners over the last decades and the level of mathematical sophistication has risen substantially. However, a mathematical model is at best a simplification of the real world phenomenon that is being modeled, and mathematical sophistication can never replace common sense and knowledge of the limitations of mathematical modeling.

Track, Mathematics of Data Science (DAVE)

The technological progress and the increased availability of information contributes to the emergence of massive and complex data sets. A variety of scientific fields are contributing to the analysis of such data at the interface of mathematics, statistics, optimization and computational methods for learning. Optimal decision making under uncertainty based in such circumstances require modelling and discovering relevant features in data, optimization of decision policies and model parameters, dimension reduction and large scale computations. Data science based on applied mathematics has the potential for transformative impact on natural sciences, business and social sciences.

Track, Optimization and Systems Theory (OPST)

Optimization and Systems Theory is a discipline in applied mathematics primarily devoted to methods of optimization, including mathematical programming and optimal control, and systems theoretic aspects of control and signal processing. The discipline is also closely related to mathematical economics and applied problems in operations research, systems engineering and control engineering.

Master's education in Optimization and Systems Theory provides knowledge and competence to handle various optimization problems, both linear and nonlinear, to build up and analyze mathematical models for various engineering systems, and to design optimal algorithms, feedback control, and filters and estimators for such systems.

Optimization and Systems Theory has wide applications in both industry and research. Examples of applications include aerospace industry, engineering industry, radiation therapy, robotics, telecommunications, and vehicles. Furthermore, many new areas in biology, medicine, energy and environment, and information and communications technology require understanding of both optimization and system integration.

Track, Computer Simulations for Science and Engineering (CSSE)

This track CSSE is given in cooperation by three European universities: Royal Institute of Technology, KTH (Sweden), Delft University of Technology, TUD (Netherlands), and Technische Universität Berlin, TUB (Germany).

The main objective of CSSE is to train students in Computational Science and Engineering (CSE) and to prepare them for international research and development employment in academy, industry, and services sector.

A student of this two-years programme studies during the first year at one of the three universities, normally TU Berlin and then continues his/her studies at another university in a different country.

The COSSE programme offers a number of specialisations within the spearhead competencies of each partner university. A student chooses his/her specialisation during the second semester and continues with courses and the degree project at the other university in this specialisation.